Lösung: Gauss-Verfahren

a)
$$\begin{aligned} 2x + 4y &= 2 \\ 8x + 4y &= 8 \end{aligned}$$

$$\Rightarrow \left(\begin{array}{cc|c} 2 & 4 & 2 \\ 8 & 4 & 8 \end{array}\right) \qquad G_2 - 4\,G_1$$

$$\Rightarrow \left(\begin{array}{cc|c} 2 & 4 & 2 \\ 0 & -12 & 0 \end{array}\right) \Rightarrow -12y = 0 \Rightarrow y = 0$$

in G_1: $2x + 4 \cdot 0 = 2 \Rightarrow 2x = 2 \Rightarrow x = 1$

Lösungsmenge: $\mathbb{L} = \{(x=1, y=0)\}$

b)
$$\begin{aligned} 2x + 4y &= 3 \\ 8x + 16y &= 12 \end{aligned}$$

$$\Rightarrow \left(\begin{array}{cc|c} 2 & 4 & 3 \\ 8 & 16 & 12 \end{array}\right) \qquad G_2 - 4\,G_1$$

$$\Rightarrow \left(\begin{array}{cc|c} 2 & 4 & 3 \\ 0 & 0 & 0 \end{array}\right) \Rightarrow$$ LGS hat unendlich viele Lösungen!

in G_1: $2x + 4y = 3 \Rightarrow x = \frac{3}{2} - 2y$

Lösungsmenge: $\mathbb{L} = \{(x = \frac{3}{2} - 2t, y = t) \mid t \in \mathbb{R}\}$

c) $\quad 2x + 4y = 3$

$\quad 8x + 16y = 6$

$$\Rightarrow \left(\begin{array}{cc|c} 2 & 4 & 3 \\ 8 & 16 & 6 \end{array}\right) \qquad G_2 - 4G_1$$

$$\Rightarrow \left(\begin{array}{cc|c} 2 & 4 & 3 \\ 0 & 0 & -6 \end{array}\right) \qquad \Rightarrow \text{ LGS hat keine Lösung}$$

Lösungsmenge: $\mathbb{L} = \{\} = \emptyset$

d) $\quad x + y + z = 42$

$\quad 2x + y - z = 21$

$\quad 5x + 2y + 3z = -21$

$$\Rightarrow \left(\begin{array}{ccc|c} 1 & 1 & 1 & 42 \\ 2 & 1 & -1 & 21 \\ 5 & 2 & 3 & -21 \end{array}\right) \qquad \begin{array}{l} G_2 - 2G_1 \\ G_3 - 5G_1 \end{array}$$

$$\Rightarrow \left(\begin{array}{ccc|c} 1 & 1 & 1 & 42 \\ 0 & -1 & -3 & -63 \\ 0 & -3 & -2 & -231 \end{array}\right) \qquad G_3 - 3G_2$$

$$\Rightarrow \left(\begin{array}{ccc|c} 1 & 1 & 1 & 42 \\ 0 & -1 & -3 & -63 \\ 0 & 0 & 7 & -42 \end{array}\right) \qquad \Rightarrow \quad 7z = -42 \quad \Rightarrow \quad z = -6$$

in G: $\quad -y - 3 \cdot (-6) = -63 \quad \Rightarrow \quad y = 63 + 18 \quad \Rightarrow \quad y = 81$

in G_1: $\quad x + 81 + (-6) = 42 \quad \Rightarrow \quad x = -33$

Lösungsmenge: $\mathbb{L} = \{(x = -33,\ y = 81,\ z = -6)\}$

e)
$$\begin{aligned} x_1 + 2x_2 + 3x_3 + x_4 &= 0 \\ x_1 + 3x_2 + 2x_3 + x_4 &= 1 \\ 2x_1 + x_2 - x_3 + 4x_4 &= 1 \\ x_1 + 6x_2 + 4x_3 + 2x_4 &= 0 \end{aligned}$$

$$\Rightarrow \left(\begin{array}{cccc|c} 1 & 2 & 3 & 1 & 0 \\ 1 & 3 & 2 & 1 & 1 \\ 2 & 1 & -1 & 4 & 1 \\ 1 & 6 & 4 & 2 & 0 \end{array}\right) \quad \begin{array}{l} G_2 - G_1 \\ G_3 - 2G_1 \\ G_4 - G_1 \end{array}$$

$$\Rightarrow \left(\begin{array}{cccc|c} 1 & 2 & 3 & 1 & 0 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & -3 & -7 & 2 & 1 \\ 0 & 4 & 1 & 1 & 0 \end{array}\right) \quad \begin{array}{l} G_3 + 3G_2 \\ G_4 - 4G_2 \end{array}$$

$$\Rightarrow \left(\begin{array}{cccc|c} 1 & 2 & 3 & 1 & 0 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & -10 & 2 & 4 \\ 0 & 0 & 5 & 1 & -4 \end{array}\right) \quad G_4 + \tfrac{1}{2} G_3$$

$$\Rightarrow \left(\begin{array}{cccc|c} 1 & 2 & 3 & 1 & 0 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & -10 & 2 & 4 \\ 0 & 0 & 0 & 2 & -2 \end{array}\right) \quad \Rightarrow 2x_4 = -2 \Rightarrow x_4 = -1$$

in G_3: $-10x_3 + 2(-1) = 4 \Rightarrow -10x_3 = 6 \; ; \; x_3 = -\tfrac{3}{5}$

in G_2: $x_2 - (-\tfrac{3}{5}) = 1 \Rightarrow x_2 = 1 - \tfrac{3}{5} \Rightarrow x_2 = \tfrac{2}{5}$

in G_1: $x_1 + 2 \cdot \tfrac{2}{5} + 3 \cdot (-\tfrac{3}{5}) - 1 = 0 \Rightarrow x_1 = 2$

Lösungsmenge: $\mathbb{L} = \{(x_1 = 2 \; ; \; x_2 = \tfrac{2}{5} \; ; \; x_3 = -\tfrac{3}{5} \; ; \; x_4 = -1)\}$

(f)

$$\begin{aligned} x_1 + 2x_2 &= 0 \\ 2x_1 + 3x_2 - 2x_3 &= 0 \\ x_1 + x_2 - 2x_3 &= 1 \end{aligned}$$

$$\Rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 2 & 3 & -2 & 0 \\ 1 & 1 & -2 & 1 \end{array}\right) \qquad \begin{array}{l} G_2 - 2G_1 \\ G_3 - G_1 \end{array}$$

$$\Rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & -1 & -2 & 0 \\ 0 & -1 & -2 & 1 \end{array}\right) \qquad G_3 - G_2$$

$$\Rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & -1 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right) \Rightarrow \text{LGS hat keine Lösung.}$$

Lösungsmenge: $\mathbb{L} = \{\} = \emptyset$

g)
$$\begin{aligned} x_1 + 2x_2 &= 0 \\ 2x_1 + 3x_2 - 2x_3 &= 0 \\ x_1 + x_2 - 2x_3 &= 0 \end{aligned}$$

$$\Rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 2 & 3 & -2 & 0 \\ 1 & 1 & -2 & 0 \end{array}\right) \qquad \begin{array}{l} G_2 - 2G_1 \\ G_3 - G_1 \end{array}$$

$$\Rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & -1 & -2 & 0 \\ 0 & -1 & -2 & 0 \end{array}\right) \qquad G_3 - G_2$$

$$\Rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & -1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \Rightarrow$$ LGS hat unendlich viele Lösungen!

u G_2: $-x_2 - 2x_3 = 0 \Rightarrow x_2 = -2x_3$

u G_1: $x_1 + 2(-2x_3) = 0 \Rightarrow x_1 = 4x_3$

Lösungsmenge: $\mathbb{L} = \{(x_1 = 4t;\ x_2 = -2t;\ x_3 = t) \mid t \in \mathbb{R}\}$